

Lecture 4: Box Model, Page Layout & Linking

4.1 Box model and Position

4.2 Multi-Column Layout

4.3 Pseudo-classes, Linking & Common Tips



{ CSS }



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Cascading Style Sheet

4.1 BOX MODEL & POSITION

CSS Box Model

Content

- *Text & web page elements in the container*

Padding

- *Area between the content and the border*

Border

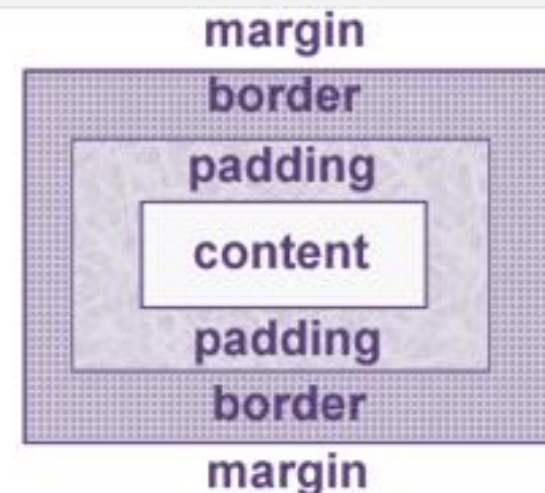
- *Between the padding and the margin*

Margin

- *Determines the empty space between the element and adjacent elements*

Syntax

```
div { width: 200px;
      padding: 100px;
      border: 25px solid blue;
      margin: 50px; }
```

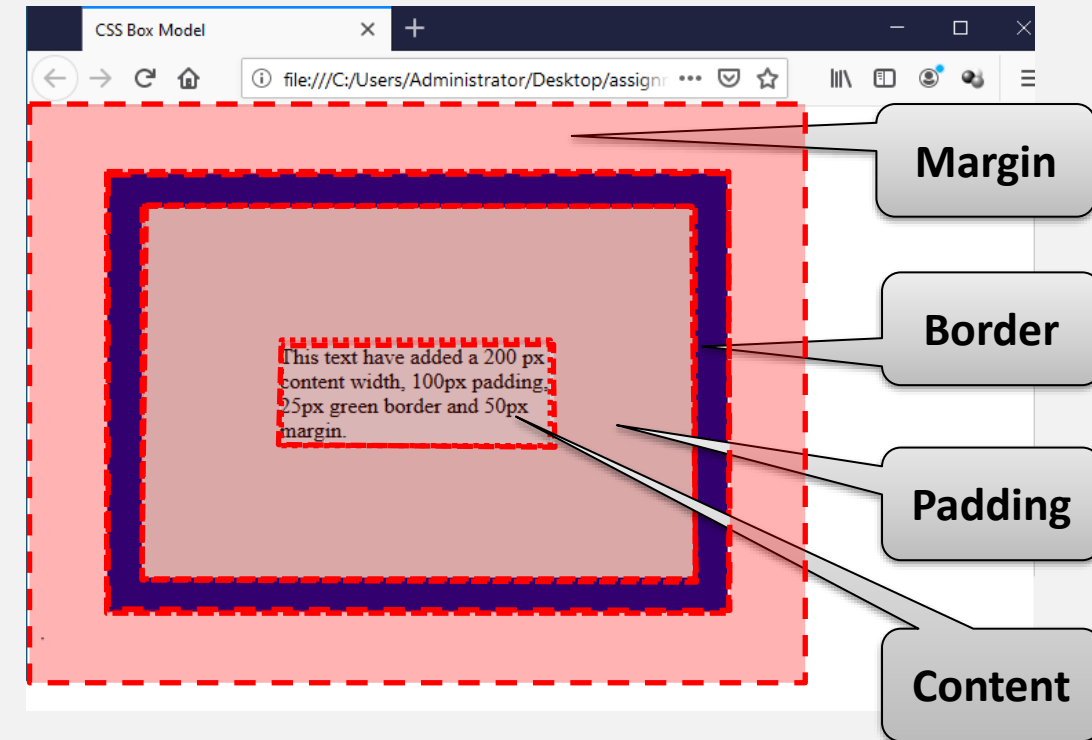


Example of Box model



```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
<head>
<title> CSS Box Model </title>
<style>
div { background-color: lightgrey;
width: 200px; padding: 100px;
border: 25px solid darkblue; margin: 50px;}
</style>
</head>
<body>
<div>This text have added a 200 px content width, 100px
padding, 25px green border and 50px margin.</div>
</body>
</html>
```



Box Model Properties

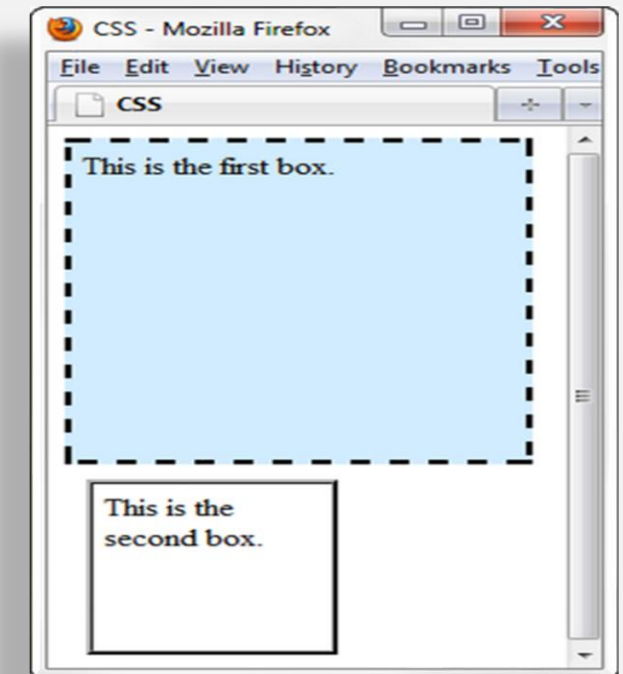


Property	Description	
- height: 300px; - width: 75%;	- To define the <i>height/ width of the content</i> inside an element.	- auto (default): calculated by browser. - length: defined in px, cm etc. - % : defined in percentage.
- padding: 30px; - padding-top: 40%; - padding-right: 50px; - padding-bottom: 60%; - padding-left: 70px;	- To generate padding space between an element's content and the border. (top, right, bottom, and left)	
- margin: 30%; - margin-top: 40 px; - margin-right: 50%; - margin-bottom: 60 px; - margin-left: 70 px;	- To specify a margin area outside the border. (top, right, bottom, and left)	
- border: 50px solid grey;	- To specify the <i>width, style and color</i> of an element's border between the padding and margin.	- style: none, dotted, dashed, solid, double, etc - color: defined color name like "red"; or HEX value like "#ff0000"; or RGB like "rgb(255,0,0)"

Normal Flow

- Browser *display of elements in the order* they are coded in the document

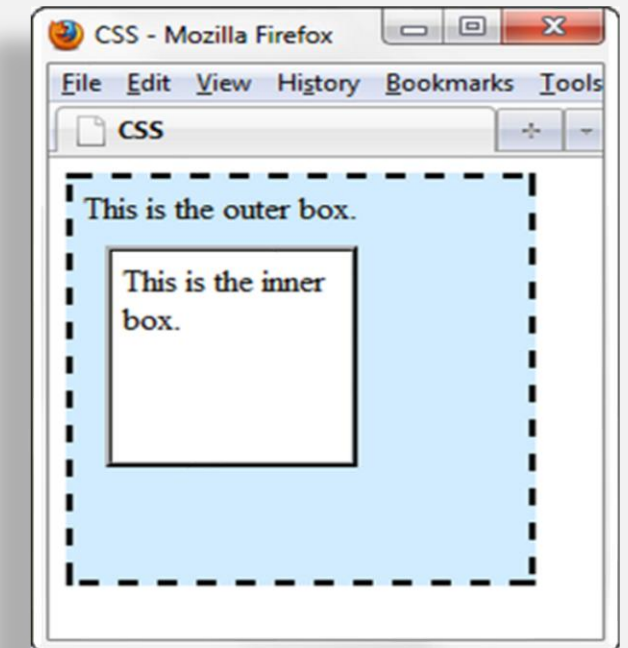
```
<html>
  <head>
    <style>
      #div1 {width: 200px; height: 200px; background-color:
      #D1ECFF; border: 3px dashed #000000; padding: 5px; }
      #div2 {width: 100px; height: 100px; background-color:
      #ffffff; border: 3px ridge #000000; padding: 5px; margin:
      10px; }
    </style>
  </head>
  <body>
    <div id="div1"> This is the outer box.</div>
    <div id="div2"> This is the inner box.</div>
  </body>
</html>
```



Iterative Flow

- To display the *element in another element*, write the code as follow:

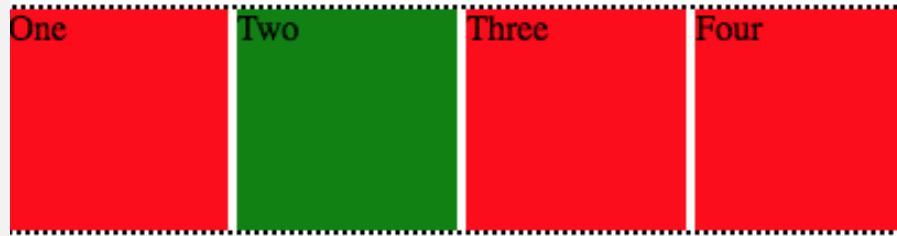
```
<html>
  <head>
    <style>
      #div1 {width: 200px; height: 200px; background-color:
      #D1ECFF; border: 3px dashed #000000; padding: 5px; }
      #div2 {width: 100px; height: 100px; background-color:
      #ffffff; border: 3px ridge #000000; padding: 5px; margin:
      10px; }
    </style>
  </head>
  <body>
    <div id="div1"> This is the outer box.
      <div id="div2"> This is the inner box.</div>
    </div>
  </body>
</html>
```



CSS Position Property

- ✓ This property specifies the *type of positioning* method for an element.

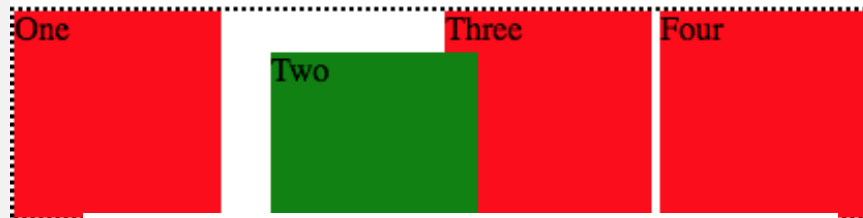
position: static



static

according to the
page normal flow

position: relative



relative

adjust the location
from its normal
position

position: absolute



absolute

positioned relative to
the nearest positioned
ancestor/ moves along
with page scrolling

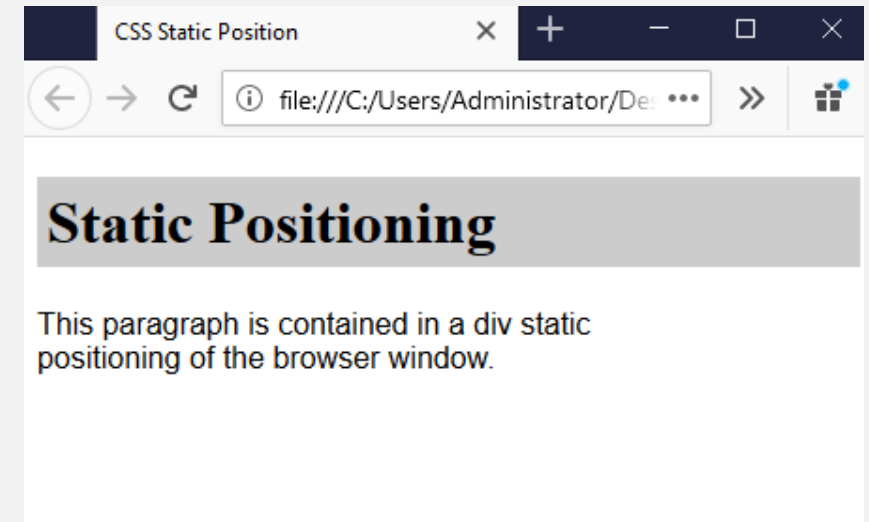
sticky

positioned based on
the user's scroll
position

Static Positioning Example



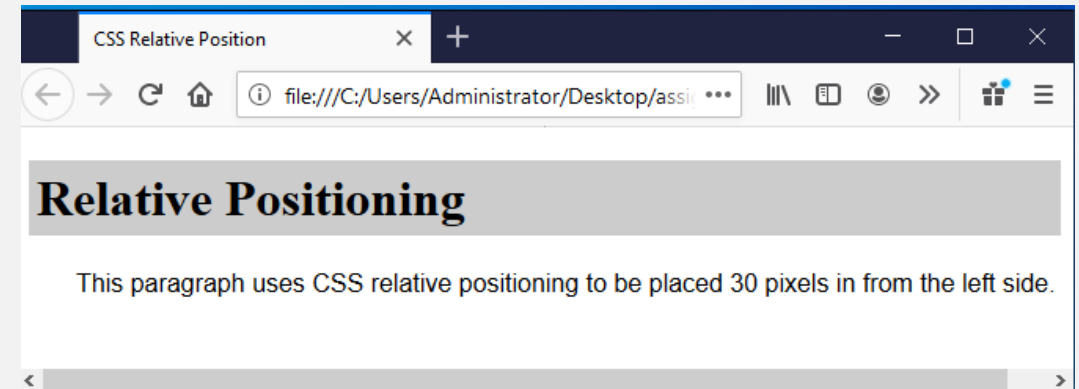
```
<!DOCTYPE html>
<html>
  <head>
    <style>
      h1 { background-color:#cccccc; padding: 5px;
        color: #000000; }
      .content {position: static;
        left: 200px; top: 100px;
        font-family: Arial,sans-serif; width:
        300px;}
    </style>
    <title>CSS Static Position</title>
  </head>
  <body>
    <h1> Static Positioning</h1>
    <div class="content">
      <p> This paragraph is contained in a div static
        positioning of the browser window.</p>
    </div>
  </body>
</html>
```



Relative Positioning Example



```
<!DOCTYPE html>
<html>
  <head>
    <style>
      h1 { background-color:#cccccc;
        padding: 5px;
        color: #000000; }
      .myContent { position: relative;
        left: 30px;
        font-family: Arial,sans-serif; }
    </style>
    <title>CSS Relative Position</title>
  </head>
  <body>
    <h1> Relative Positioning </h1>
    <div class="myContent">
      <p> This paragraph uses CSS relative
        positioning to be placed 30 pixels in from the
        left side.</p>
    </div>
  </body>
</html>
```





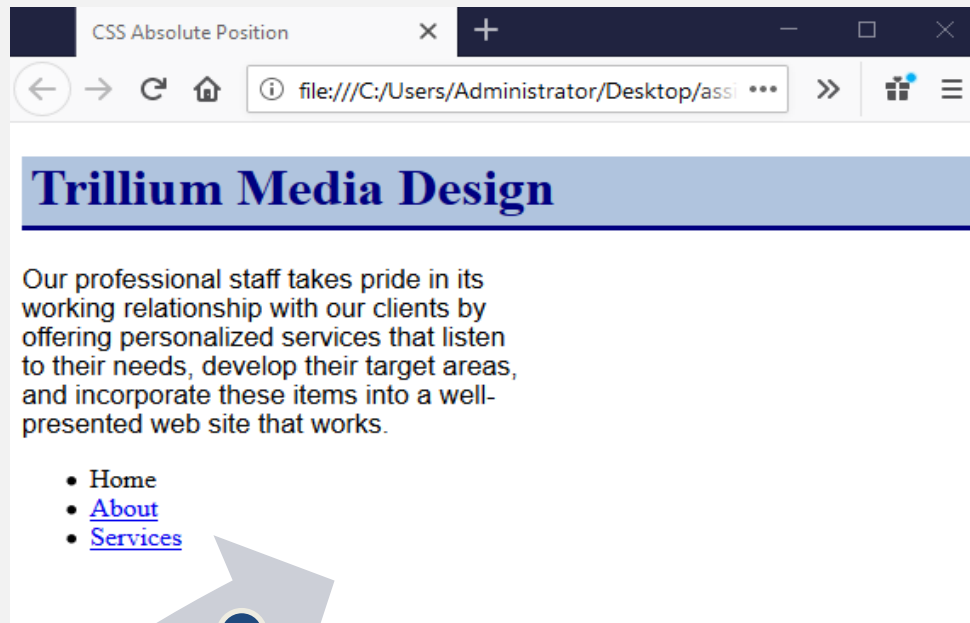
Absolute Positioning Example: Trillium Media Design

```
<!DOCTYPE html>
<html>
<head> <style>
h1 { border-bottom: 3px solid #000080; color: #000080;
background-color: #B0C4DE; padding: 0 0 5px 5px; }
#content { position: absolute; left: 200px; top: 75px; font-
family: Arial, sans-serif; width: 300px; }
</style> </head>
<body>
<h1>Trillium Media Design</h1>
<div>
<p id="content">Our professional staff takes pride in its
working relationship with our clients by offering personalized
services that listen to their needs, develop their target
areas, and incorporate these items into a well-presented web
site that works. </p> </div>
<ul> <li>Home</li>
<li><a href="about.html">About</a> </li>
<li><a href="services.html"> Services</a></li> </ul>
</body>
</html>
```

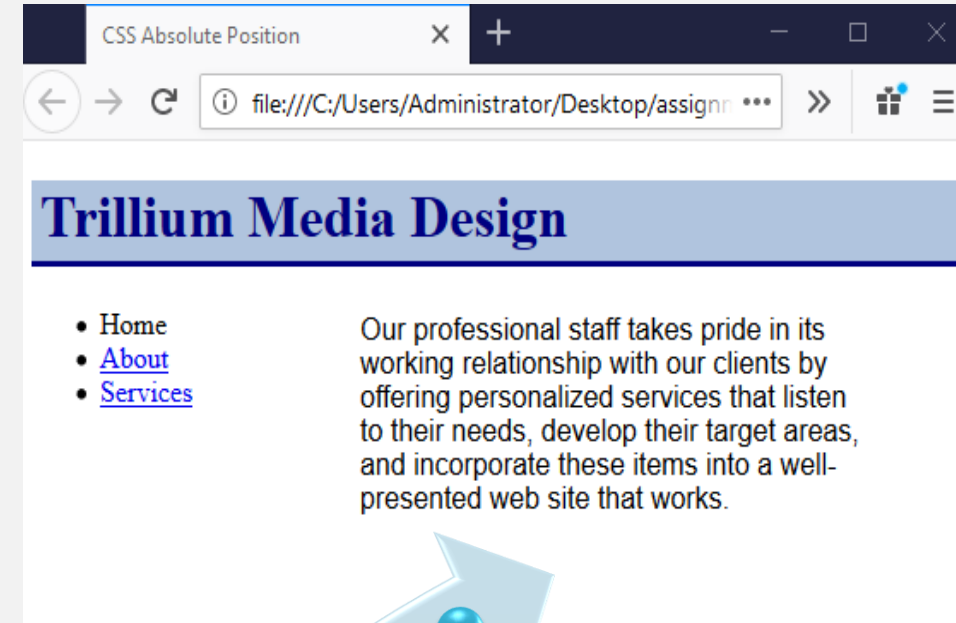


How to use Absolute Positioning???

position



Without
absolute
position



With
absolute
position

Lets practice!



According to the previous CSS
Position example,

- Write the code for different
position styles of CSS.



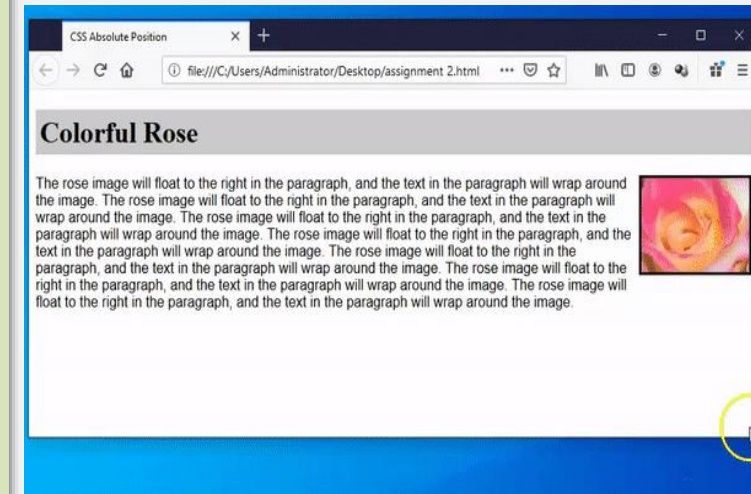
Float Property Example: Colorful rose



Syntax

```
div { float: none|left|right; }
```

```
<!DOCTYPE html>
<html> <head> <style>
  h1 { background-color:#cccccc; padding:5px; color: #000000;
    }
  p { font-family:Arial,sans-serif; }
  #yls {float:right; margin: 0 0 5px 5px; border: solid;
    }</style>
</head>
<body> <h1>Colorful Rose</h1>
<p> The rose image will float to the right in the
paragraph, and the text in the paragraph will wrap around the image.
The rose image will float to the right in the paragraph, and the text
in the paragraph will wrap around the image. The rose image will float
to the right in the paragraph, and the text in the paragraph will wrap
around the image. The rose image will float to the right in the
paragraph, and the text in the paragraph will wrap around the image.
The rose image will float to the right in the paragraph, and the text
in the paragraph will wrap around the image. </p>
</body> </html>
```



CSS clear property

When an element is floated, it will get out of the normal layout rules of document.

CSS {clear: left | right | both}
applies after floating element.

So, it is useful to “clear” or *terminate another element float to its left or right.*

Syntax

```
■ div {  
    clear: left | right | both;  
}
```

With/ Without using clear property

```
<!DOCTYPE html>
<html>
<head>
  <style>
    #div1{ float: right; width: 100px;
    height: 50px; margin: 10px; border: 3px;
    solid #AD7321; }
    #div2{ border: 1px solid red; }
    #div3{ float: right; width: 100px;
    height: 50px; margin: 10px; border: 3px
    solid #AD7321; }
    #div4{ border: 1px solid red; clear:
    right;}
  </style>
  <title>With/ Without clear property
</title></head>
<body>
  <h2>Without clear property</h2>
```

```

<div id="div2"> In the code, be carefully
aware that div2 is declared after div1.
However, as div1 floats to the right, the
text in div2 element flows around
div1.</div>
<br><br>
<h2>With clear property</h2>

<div id="div4">In this code, clear: right;
moves div4 down below the floating div3.
The value "clear:right" clears elements
floated to the right. You can also test
"clear:left" and "clear:both".</div>
</body> </html>
```


CSS clear property



With/ Without clear property

file:///C:/Users/Administrator/Desktop/assi

Without clear property

In the code, be carefully aware that div2 is declared after div1. However, as div1 floats to the right, the text in div2 element flows around div1.



With clear property



In this code, `clear: right;` moves div4 down below the floating div3. The value `"clear:right"` clears elements floated to the right. You can also test `"clear:left"` and `"clear:both"`.

Overflow Property

- Specifies *whether to add scrollbars* when an element's content is big to fit in specified area.
- Values are auto, hidden, and scroll.

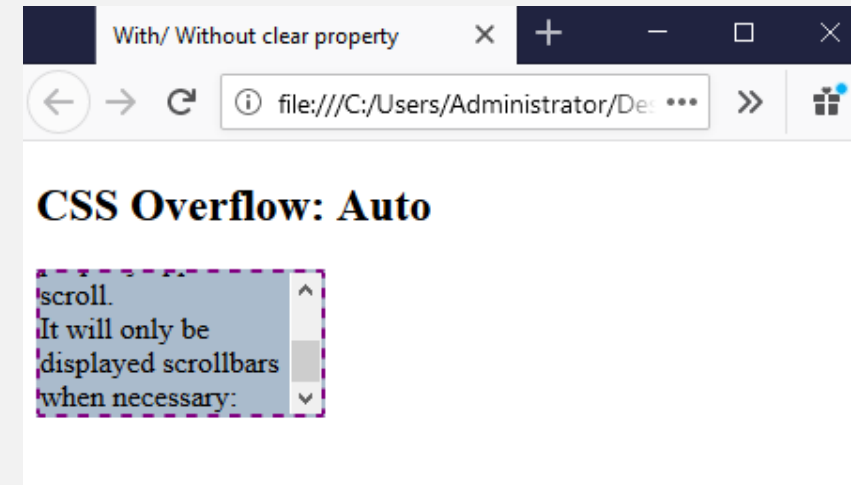
Syntax

- ```
div {
 overflow: visible | hidden | scroll | auto;
}
```

# Example using overflow property

```
<!DOCTYPE html>
<html>
<head>
 <style>
 div {
 background-color:#abc; width:150px; height:75px;
 border:2px dashed purple; overflow:auto; }
 </style>
</head>
<body>
 <h2>CSS Overflow: Auto</h2>
 <div>The auto value in CSS overflow property appears as
 scroll.

It will only be displayed scrollbars when necessary:
 </div>
</body>
</html>
```



# Display Property

- Configures how and if an element is displayed
  - **display: none;**
    - The element will **not be displayed**.
  - **display: block;**
    - The element is rendered **as a block element** – even if it is an inline element, such as a hyperlink `<a>`, `<span>`, `<img>`.
  - **display: inline;**
    - The element will be rendered **as an inline element** – even if it is a block element – such as a `<li>`, `<div>`, `<h1> ~ <h6>`, `<p>`, `<header>`, `<footer>`, etc.

## Lets practice!



According to the CSS float and clear property example,

- Write the code for CSS float property, and
- test “with or without CSS float property” with left|right| both values.



# Summary

## Lecture 4.1: Box model and Positioning

- ✓ CSS Box Model
- ✓ Normal Flow
- ✓ CSS Positioning
- ✓ CSS Properties
  - Float
  - Clear
  - Overflow
  - Display



# { CSS }



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## Cascading Style Sheet

### 4.2 MULTI-COLUMN LAYOUT

# Multi Column Property

- The CSS multi-column layout allows multiple columns of text.
- The **column-count** property specifies the number of columns and an element will be divided according to it.

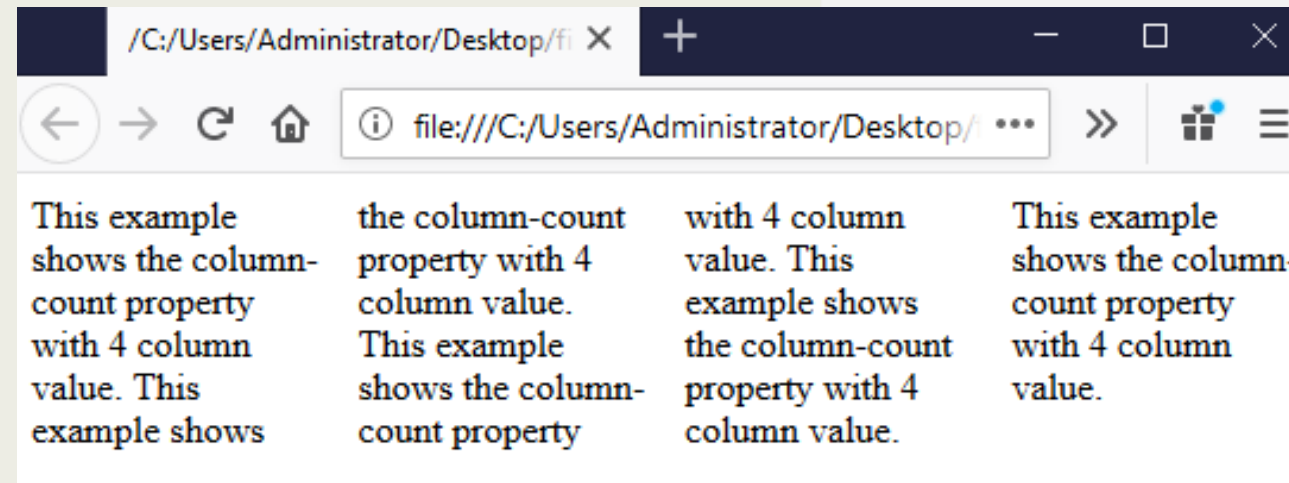
## Syntax

```
• div {
 column-count: 3;
}
```



# Example of Multi Column Property

```
<!DOCTYPE html>
<html>
<head>
<style>
#cc { column-count: 4; }
</style>
</head>
<body>
<div id="cc">
This example shows the column-count property with 4 column values.
This example shows the column-count property with 4 column values.
This example shows the column-count property with 4 column values.
This example shows the column-count property with 4 column values.
This example shows the column-count property with 4 column values.
</div>
</body>
</html>
```

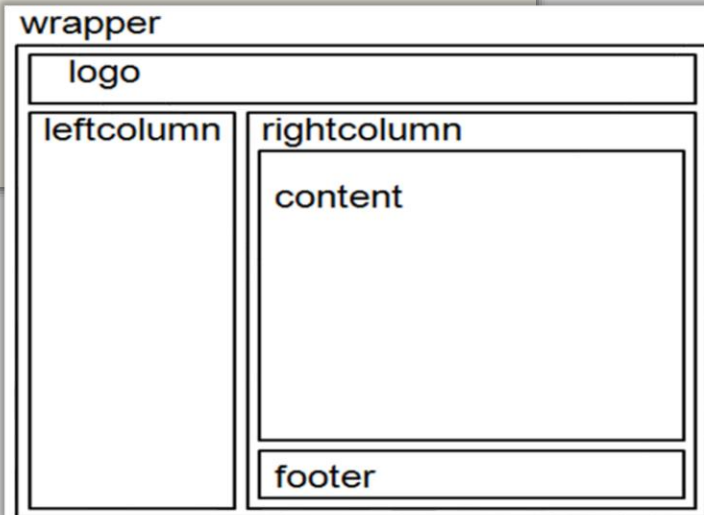




# Multi-Column with Navigation

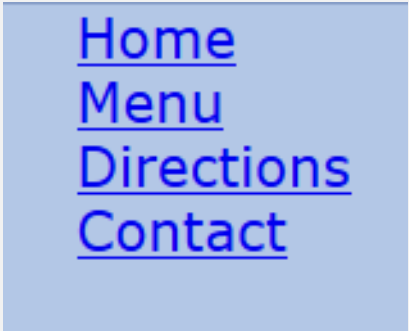
# Example of Two Columns (top logo, left nav)

```
<div id="wrapper">
 <div id="header">..
 <div id="leftcolumn">..
 <div id="rightcolumn">
 <div id="content">..
 <div id="footer">..
 </div>
</div>
```



# Navigation Element <nav>

- contains a section of navigation hyperlinks
- block display



[Home](#)  
[Menu](#)  
[Directions](#)  
[Contact](#)

## Syntax

<nav>

<a href="index.html">...</a>

<a href="rooms.html">...</a>

<a href="contact.html">...</a>

</nav>

# Example of Two Columns (top logo, left nav)

twocolumn.html

```
<!DOCTYPE html>
<html>
 <head>
 <title> Lighthouse Island Bistro </title>
 <link rel="stylesheet" href="twocolumn.css">
 </head>
 <body>
 <div id="wrapper">
 <header><h1>Lighthouse Island Bistro</h1></header>
 <div id="leftcol">
 <nav>
 Home
 Menu
 Directions
 Contact
 </nav>
 </div>
 <div id="rightcol">

```

```
 <h2>Locally Roasted Free-Trade Coffee</h2>
 <p>Indulge in the aroma of freshly ground roast
 coffee. Specialty drinks are available hot or
 cold.</p>
 <h2>Specialty Pastries</h2>
 <p>Enjoy a selection of our fresh-baked, organic
 pastries, including fresh-fruit muffins, scones,
 croissants, and cinammon rolls.</p>
 <h2>Lunchtime is Anytime</h2>
 <p>Savor delicious wraps and sandwiches on
 hearty, whole-grain breads with locally-grown
 salad, fruit, and vegetables. </p>
 <h2>Panoramic View</h2>
 <p>Take in some scenery! The top of our
 lighthouse offers a panoramic view of the
 countryside. Challenge your friends to climb our
 100-stair tower.</p>
 <footer>Copyright © 2014</footer>
 </div></div>
 </body>
</html>
```

# Example of Two Columns (top logo, left nav)



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twocolumn.css

```
body {font-family:Verdana, Arial, sans-serif;
background-color: #00005D; }

#wrapper { background-color: #b3c7e6; width: 80%;
margin: auto; min-width: 800px; max-width: 1200px; }

nav { float: left; width: 150px; }

#leftcol { margin-left: 15px; background-color:
#000000; color: #ffffff; }

#rightcol { margin-left: 155px; background-color:
#ffffff; color: #000000; }

header { background-color: #869dc7; color: #00005D;
font-size: 150%; padding: 30px; }

h2 { color: #869dc7; font-family: arial, sans-serif;
}

#floatright { margin: 10px; float: right; }

footer { font-size:70%; text-align: center; clear:
right; padding-bottom:20px; }

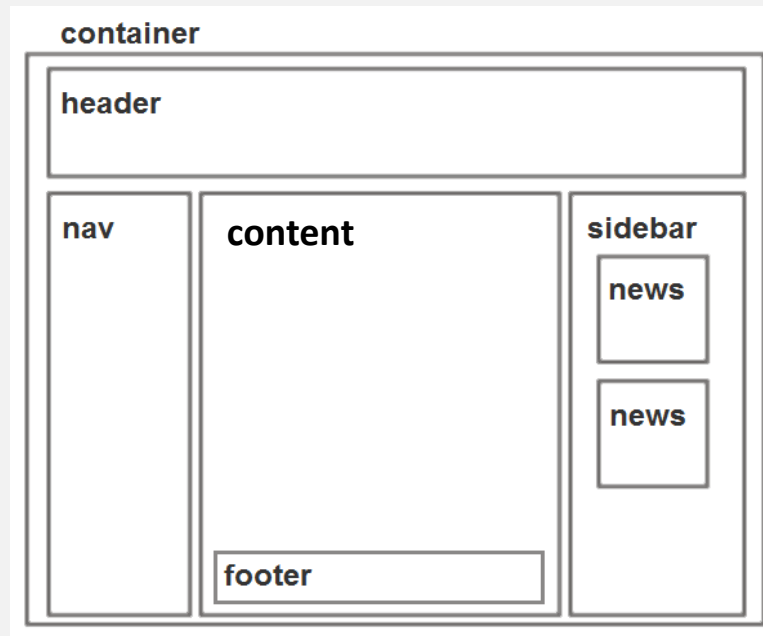
a { margin: 30px; }
```



# Three Column Page Layout

A common web page layout consists of a header across the top of the page with three columns below:

1. navigation,
2. content, and
3. sidebar.



# Lets practice!

According to the two column example, write three column layout using the following tips:

- Container sets default background color, text color, font typeface, and a minimum width
- **Left-column navigation**-- `float: left;` width:150px;
- **Right-column content**-- `float: right;` width: 200px;
- **Center column**-- Uses the remaining screen room available room after the floating columns display
  - `margin: 0 210px 0 160px;`
- **Footer** – clears the float
  - `clear: both;`







# Navigation Hyperlink Configuration

# Configure Hyperlinks in an Unordered List

## ■ Vertical Navigation



```
<div id="nav">

 Home
 Menu
 Directions
 Contact

</div>
```

# Configure Hyperlinks in an Unordered List (cont'd)

## CSS can ...

- remove the [list marker](#),
- remove the [underline](#),

```
#nav ul { list-style-type: none; }
#nav a { text-decoration: none; }
```



Home  
Menu  
Directions  
Contact

# Configure Hyperlinks in an Unordered List (cont'd)

## Horizontal Navigation

Home Menu Directions Contact

```
<div id="nav">

 Home
 Menu
 Directions
 Contact

</div>
```

```
#nav a { padding-right: 10px;}
#nav li{ display: inline;}
```

## CSS can ...

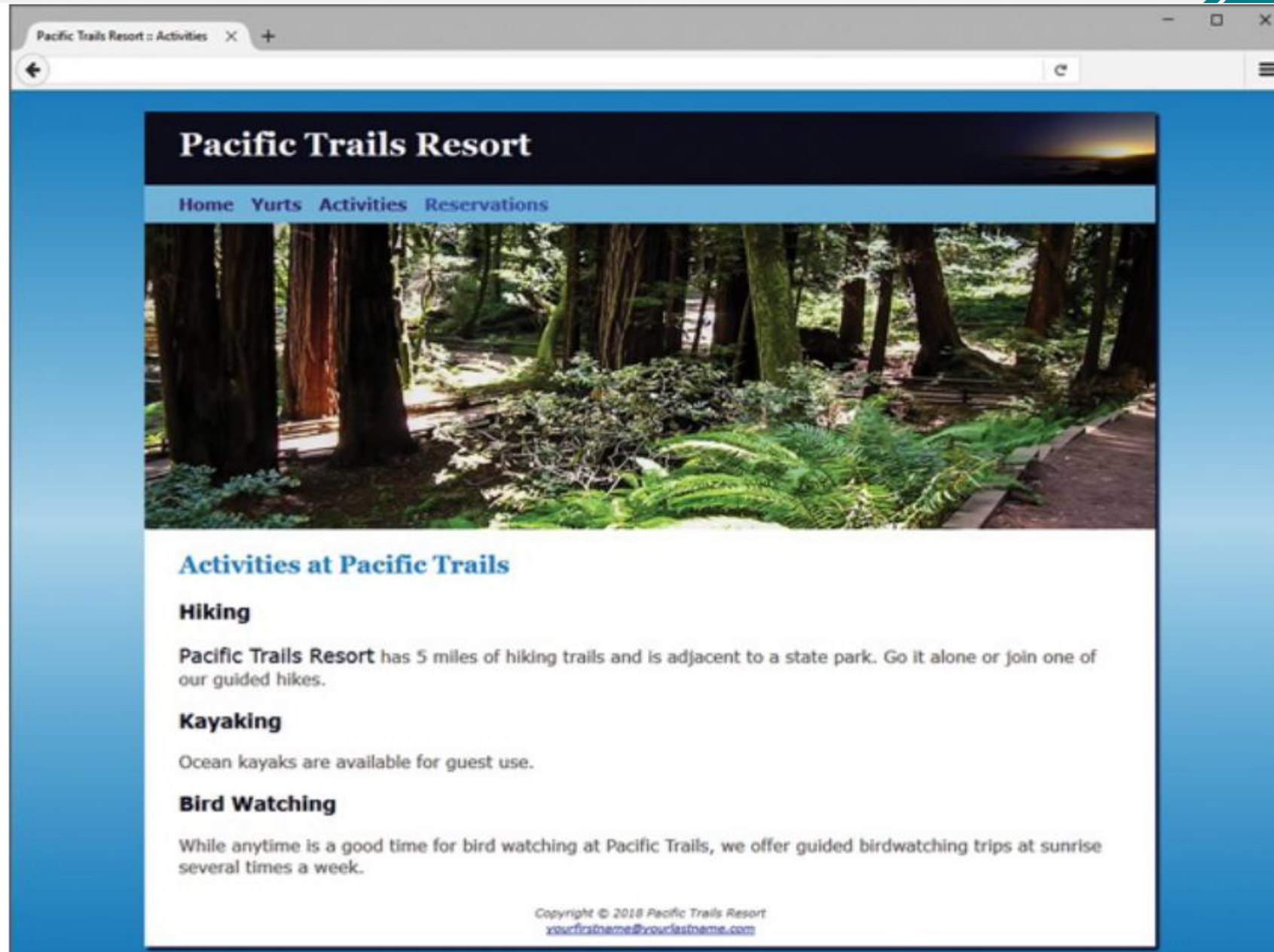
- add **padding**, and
- configure the list items for **inline** display.

# Summary

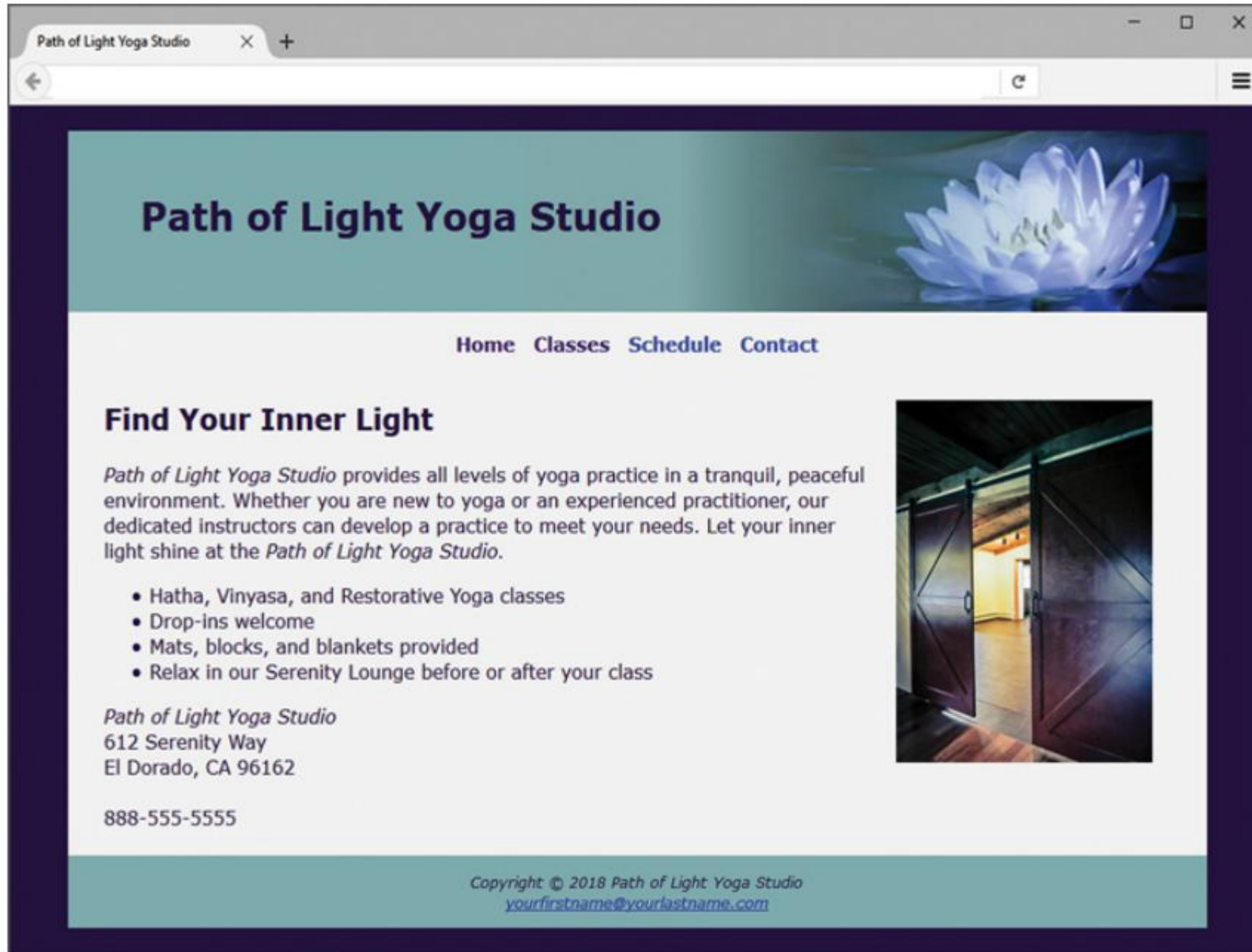
## Lecture 4.2: MULTI-Column Layout

- ✓ Multi-Column Property
- ✓ Multi-Column with Navigation
- ✓ Navigation Hyperlink Configuration









# Path of Light Yoga Studio



[Home](#) [Classes](#) [Schedule](#) [Contact](#)



## Yoga Classes

### Gentle Hatha Yoga

Intended for beginners and anyone wishing a grounded foundation in the practice of yoga, this 60 minute class of poses and slow movement focuses on asana (proper alignment and posture), pranayama (breath work), and guided meditation to foster your mind and body connection.

### Vinyasa Yoga

Although designed for intermediate to advanced students, beginners are welcome to sample this 60 minute class that focuses on breath-synchronized movement — you will inhale and exhale as you flow energetically



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# { CSS }



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## Cascading Style Sheet

### 4.3 PSEUDO-CLASSES, LINKING, COMMON TIPS

# Structural Elements

Header Element

contains the  
**headings** of **either a  
web page document  
or an area in the  
document.**

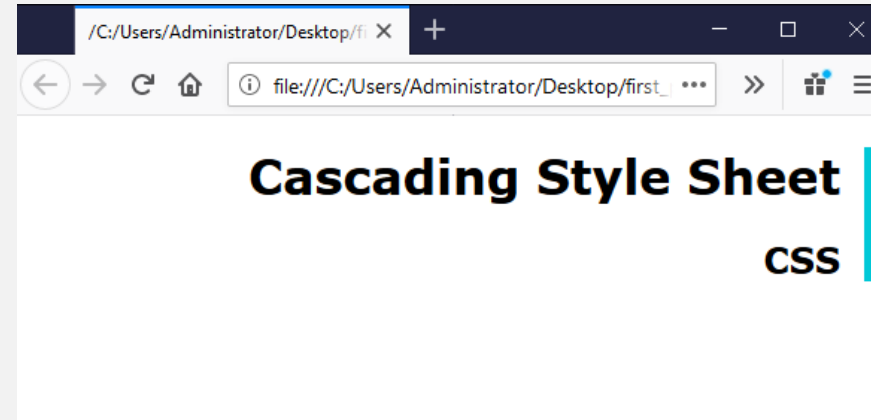
Hgroup Element

contains **two or  
more heading  
elements** (h1, h2,  
and so on)

# Header and Hgroup Structural Elements



```
<!DOCTYPE html>
<html>
<head>
<style>
hgroup { text-align: right; font-
family: Verdana, sans-serif;
padding-right: 16px; border-
right: 10px solid #00c8d7;}
</style>
</head>
<body>
<hgroup>
<h1>Cascading Style Sheet</h1>
<h2>CSS</h2>
</hgroup>
</body>
</html>
```





# Pseudo Classes

# CSS Pseudo-classes

A pseudo-class is used to define a special state of an element.

For example, visited and unvisited links color are different

Name	Description	Example
<b>link</b>	default state for a hyperlink	 <b>a:link</b> {color:#000066;}
<b>visited</b>	a hyperlink that has been visited	<b>a:visited</b> {color:#003366;}
<b>focus</b>	triggered when the hyperlink has focus	<b>a:focus</b> {color:#FF0000;}
<b>hover</b>	triggered when the mouse moves over the hyperlink	<b>a:hover</b> {color:#0099CC;}
<b>active</b>	triggered when the hyperlink is being clicked	<b>a:active</b> {color:#FF0000;}

# Example of CSS Pseudo-classes

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
a:link { color: red; }
```

```
a:visited { color: green; }
```

```
a:hover { color: hotpink; }
```

```
a:active { color: blue; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p>
```

```

```

```
Click me and check my colors.
```

```
 </p>
```

```
</body>
```

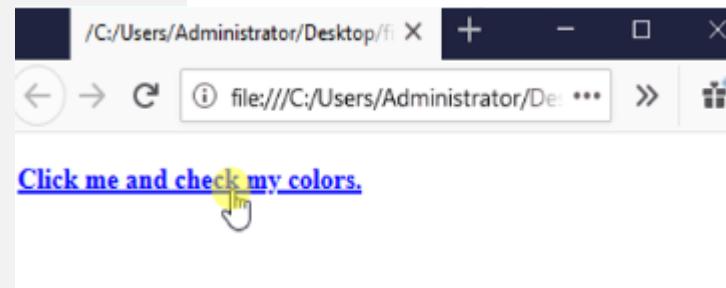
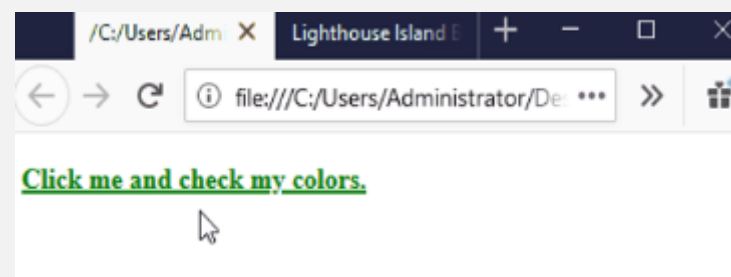
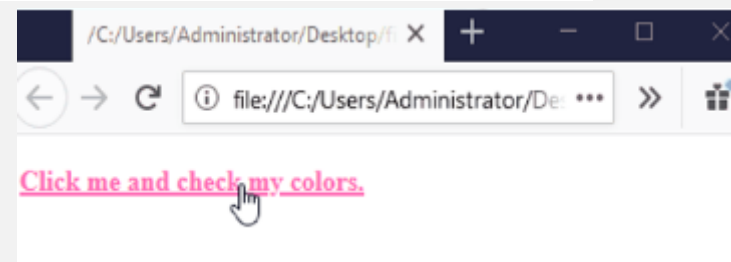
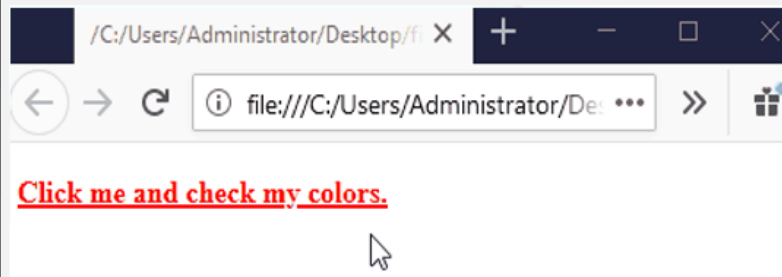
```
</html>
```

unvisited  
<a> color

visited  
<a> color

mouse over  
<a> color

selected  
<a> color



A large, light blue arrow pointing to the right, with a white outline and a slight 3D effect. It is positioned horizontally across the middle of the slide.

## More on Linking

# Opening a Link in a New Browser Window

- The **target** attribute on the anchor element opens a link in a **new browser window** or **new browser tab**.

```
<a href="http://yahoo.com"
target="_blank">Yahoo!
```

- \_blank** Opens the linked document in a **new window or tab**
- \_self** Opens the linked document in the **same frame (default)**
- \_top** Opens the linked document in the **full body of the window**



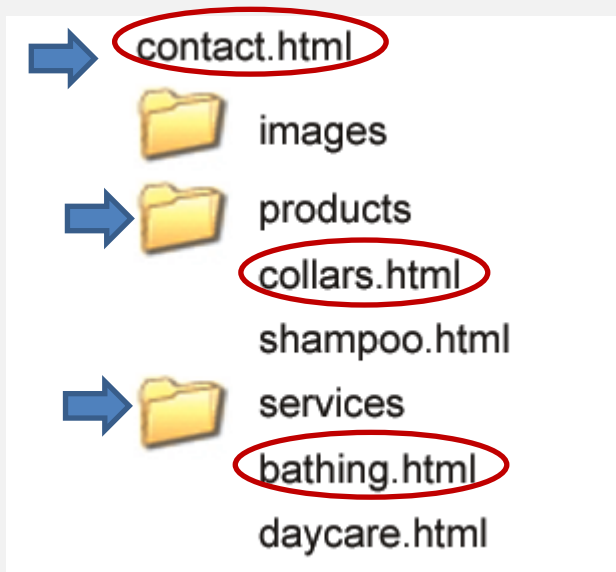
# File Path Linking

- describe location of a file in a web site's folder structure.

```
Contact
```

```
Collars
```

```
Dog Bathing
```



# HTML Linking to Fragment Identifiers

- A link to a part of a web page, also called named fragments
- Two components:
  1. The **element** that identifies the named fragment with **id** attribute:  
`<div id="top"> .... </div>`
  2. The **anchor tag** that links to the named fragment with **href** attribute:  
`<a href="#top">Back to Top</a>`

❖ *Note: the use of the # in the anchor tag!*

# HTML Linking to Fragment Identifiers

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
<h1>HTML Linking to Fragment Identifiers </h1>
<div id="top">
This is a paragraph.

This is a paragraph.

This is a paragraph.

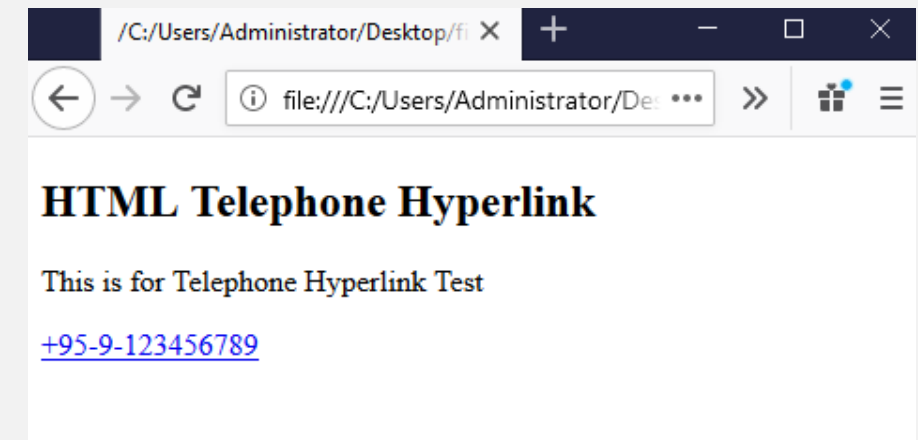
This is a paragraph.

This is a paragraph.

</div>
Go to Top
</body>
</html>
```

# Telephone & Text Message Hyperlinks

Type	Description	Example
<b>Telephone Scheme</b>	Many mobile browsers will initiate a phone call when the hyperlink is clicked.	<code>&lt;a href="tel:+95-9-123456789"&gt;Call +95-9-123456789 &lt;/a&gt;</code>
<b>SMS Scheme</b>	Many mobile browsers will initiate a text message to the phone number when the hyperlink is clicked.	<code>&lt;a href="sms:+95-9-123456789"&gt;Call +95-9-123456789 &lt;/a&gt;</code>





# Common CSS Tips

# Deciding to configure a *class* or *id*?

id

Configure an **id**:

If the style is specific to **only one element** on a page

Use the **# notation** in the style sheet.

Use the **id** attribute in the HTML.

class

Configure a **class**:

If the style may apply to **more than one element** on a page

Use the **. (dot) notation** in the style sheet.

Use the **class** attribute in the HTML.

**Note:** A class or id name  
should be descriptive of the  
purpose

# CSS Debugging Tips

- Manually check syntax errors if needed
- Use W3C CSS Validator to check syntax errors
  - <http://jigsaw.w3.org/css-validator/>
- Configure temporary background colors
- Configure temporary borders
- Use **CSS comments** to find the unexpected  
/\* the browser ignores this code \*/
- Don't expect your pages to look exactly the same in **all browsers!**
- Be patient !!!

# Summary

## Lecture 4.3: Pseudo-classes, Common Tips & Linking

- ✓ Structural Elements
- ✓ CSS Pseudo-classes
- ✓ Linking
  - ✓ File Path Linking
  - ✓ Linking to Fragment Identifiers
- ✓ CSS Common Tips

